

NAME

Maitreya Gupta



Personal

DOB: 02.10.2000
Hobbies: Karate, KickBoxing,
Game Dev, Coding
Languages: Hindi, English,
Bengali

Technical Skills

Programming: Python, C++,
Java, C
Web: React, Flask, Django,
HTML/CSS
Robotics: ROS, ESP32,
MediaPipe, YOLO
Tools: Git, VS Code, Godot,
Three.js

Social Links

- GitHub
- HackerRank
- TryHackMe Profile
- Linkedin
- Portfolio Website

Achievements

- State-level Bronze Winner in KickBoxing
- Gold Winner in Ju Srijan (Tech Fest of Jadavpur University)

OBJECTIVE

M.Tech student in Intelligent Automation & Robotics with strong expertise in Computer Vision, Robotics, Full-Stack Development, and Interactive Simulations. Seeking AI/ML, Robotics, and Software Engineering roles where I can apply my research and development skills.

EDUCATION

2023–Present	M.Tech (Intelligent Automation and Robotics) , Jadavpur University CGPA: 8.67/10, YGP: 82.75%
2019–2023	B.Tech (Electronics and Communication) , St. Thomas College CGPA: 8.33/10
2017–2019	Patha Bhavan (12th: 65%)
2015–2017	Patha Bhavan (10th: 83.28%)

SELECTED CERTIFICATIONS

- Fullstack Development - Coding Ninjas
- Python (Basic) - HackerRank
- SQL (Basic) - HackerRank
- BSNL Telecom Training

TECHNICAL ASSESSMENTS

- **TCS iON National Qualifier Test (NQT) 2026**
 - Cognitive Ability: 82.85%
 - Advanced Quantitative & Reasoning: 93.97%
 - Programming (Python): 72.76%
 - Official Scorecard

PROJECTS

Robotics & AI

- **Gesture-Controlled Robotic Arm System** — *Research, Ongoing* GitHub
 - Real-time gesture-to-actuation pipeline using MediaPipe and YOLO for 6-DOF robotic arm control.
 - Developed **CNN Regressive Motion Stabilization Filter (CRMSF)** — a novel deep learning filter for gesture jitter suppression.
 - ROS/Gazebo simulation environment; manuscript under preparation.
- **3D Medical Image Segmentation — MobiUNet3D** — *Research Live Demo*
 - Lightweight 3D segmentation architecture combining MobileNet encoder with UNet3D decoder.
 - Optimized for resource-constrained hardware without compromising accuracy.
 - Manuscript under review (2025).
- **Robotics Car Project** Drive
 - IR sensor-based obstacle detection with real-time data streaming to backend.
 - Godot Engine interface with Python for data visualization and processing.

Game Development & Simulation

- **Industrial Multi-Agent Robot Navigation Simulator** — Unity 6, C# Demo Video
 - Server-authoritative multiplayer simulator with A*, Dijkstra, RRT, and NavMesh navigation.
 - Supports Python-trained PyTorch models for custom navigation behaviors.
 - Full multiplayer synchronization with Netcode for GameObjects.
- **Inception Battlegrounds** — Roblox Studio, Lua Play
 - Multiplayer combat experience with custom gameplay systems, progression, and UI.
 - Full lifecycle development from design to deployment.

Full-Stack & Mobile Development

- **Android Job Posting Platform** — Java, Kotlin, MERN Stack Showcase
 - Full-stack job platform with secure REST APIs using Node.js, Express, and MongoDB.
 - CSV bulk upload for efficient job posting management.
 - Responsive Android UI with seamless client-server integration.

BACHELORS FINAL YEAR PROJECT

- **Compact Wideband MIMO Antenna for 5G and IoT** Report
 - Designed and simulated a compact four-element octagonal microstrip MIMO antenna using **CST Studio Suite**.
 - Achieved **2.64–20 GHz** bandwidth, **>20 dB isolation**, **6.53 dB peak gain**, and **ECC < 0.01**.
 - Fabricated and validated prototype in an anechoic chamber.

Maitreya Gupta ✉ maitreyaguptaa@gmail.com 📍 Kolkata, India ☎ +91-8697539102